

```
import pandas as pd
import numpy as np
```

```
df = pd.read_csv('PS/train.csv')
```

```
suspect_cols = [
    'Kharif Seasons Agro Ecological Sub Zone in 2022',
    'Rabi Seasons Agro Ecological Sub Zone in 2022',
    'Rabi Seasons Agro Ecological Sub Zone in 2021',
    'Kharif Seasons Agro Ecological Sub Zone in 2021',
    'Kharif Seasons Agro Ecological Sub Zone in 2020',
    'Rabi Seasons Agro Ecological Sub Zone in 2020'
]
```

```
string_correlation_matrix = df[suspect_cols].apply(lambda x: x.astype('category').cat.codes).c
print(string_correlation_matrix)
```

2022 \	Kharif Seasons Agro Ecological Sub Zone in	
Kharif Seasons Agro Ecological Sub Zone in 2022 00000		1.0
Rabi Seasons Agro Ecological Sub Zone in 2022 00000		1.0
Rabi Seasons Agro Ecological Sub Zone in 2021 00000		1.0
Kharif Seasons Agro Ecological Sub Zone in 2021 00000		1.0
Kharif Seasons Agro Ecological Sub Zone in 2020 04549		0.9
Rabi Seasons Agro Ecological Sub Zone in 2020 00000		1.0

22 \	Rabi Seasons Agro Ecological Sub Zone in 20	
Kharif Seasons Agro Ecological Sub Zone in 2022 00		1.0000
Rabi Seasons Agro Ecological Sub Zone in 2022 00		1.0000
Rabi Seasons Agro Ecological Sub Zone in 2021 00		1.0000
Kharif Seasons Agro Ecological Sub Zone in 2021 00		1.0000
Kharif Seasons Agro Ecological Sub Zone in 2020 49		0.9045
Rabi Seasons Agro Ecological Sub Zone in 2020 00		1.0000

21 \	Rabi Seasons Agro Ecological Sub Zone in 20	
Kharif Seasons Agro Ecological Sub Zone in 2022 00		1.0000
Rabi Seasons Agro Ecological Sub Zone in 2022 00		1.0000
Rabi Seasons Agro Ecological Sub Zone in 2021 00		1.0000
Kharif Seasons Agro Ecological Sub Zone in 2021 00		1.0000
Kharif Seasons Agro Ecological Sub Zone in 2020 49		0.9045
Rabi Seasons Agro Ecological Sub Zone in 2020 00		1.0000

2021 \	Kharif Seasons Agro Ecological Sub Zone in	
Kharif Seasons Agro Ecological Sub Zone in 2022 0000		1.00
Rabi Seasons Agro Ecological Sub Zone in 2022 0000		1.00
Rabi Seasons Agro Ecological Sub Zone in 2021 0000		1.00
Kharif Seasons Agro Ecological Sub Zone in 2021 0000		1.00
Kharif Seasons Agro Ecological Sub Zone in 2020		0.90

4549	
Rabi Seasons Agro Ecological Sub Zone in 2020	1.00
0000	
	Kharif Seasons Agro Ecological Sub Zone in
2020 \	
Kharif Seasons Agro Ecological Sub Zone in 2022	0.90
4549	
Rabi Seasons Agro Ecological Sub Zone in 2022	0.90
4549	
Rabi Seasons Agro Ecological Sub Zone in 2021	0.90
4549	
Kharif Seasons Agro Ecological Sub Zone in 2021	0.90
4549	
Kharif Seasons Agro Ecological Sub Zone in 2020	1.00
0000	
Rabi Seasons Agro Ecological Sub Zone in 2020	0.90
4549	
	Rabi Seasons Agro Ecological Sub Zone in 20
20	
Kharif Seasons Agro Ecological Sub Zone in 2022	1.0000
00	
Rabi Seasons Agro Ecological Sub Zone in 2022	1.0000
00	
Rabi Seasons Agro Ecological Sub Zone in 2021	1.0000
00	
Kharif Seasons Agro Ecological Sub Zone in 2021	1.0000
00	
Kharif Seasons Agro Ecological Sub Zone in 2020	0.9045
49	
Rabi Seasons Agro Ecological Sub Zone in 2020	1.0000
00	

```

suspect_cols = [
    'Kharif Seasons Type of soil in 2022',
    'Rabi Seasons Type of soil in 2022',
    'Rabi Seasons Type of soil in 2021',
    'Kharif Seasons Type of soil in 2021',
    'Kharif Seasons Type of soil in 2020',
    'Rabi Seasons Type of soil in 2020'
]

string_correlation_matrix = df[suspect_cols].apply(lambda x: x.astype('category').cat.codes).c

print(string_correlation_matrix)

```

	Kharif Seasons	Type of soil in 2022	\
Kharif Seasons	Type of soil in 2022	1.000000	
Rabi Seasons	Type of soil in 2022	1.000000	
Rabi Seasons	Type of soil in 2021	1.000000	
Kharif Seasons	Type of soil in 2021	1.000000	
Kharif Seasons	Type of soil in 2020	0.896694	
Rabi Seasons	Type of soil in 2020	1.000000	

	Rabi Seasons	Type of soil in 2022	\
Kharif Seasons	Type of soil in 2022	1.000000	
Rabi Seasons	Type of soil in 2022	1.000000	
Rabi Seasons	Type of soil in 2021	1.000000	
Kharif Seasons	Type of soil in 2021	1.000000	
Kharif Seasons	Type of soil in 2020	0.896694	
Rabi Seasons	Type of soil in 2020	1.000000	

	Rabi Seasons	Type of soil in 2021	\
Kharif Seasons	Type of soil in 2022	1.000000	
Rabi Seasons	Type of soil in 2022	1.000000	
Rabi Seasons	Type of soil in 2021	1.000000	
Kharif Seasons	Type of soil in 2021	1.000000	
Kharif Seasons	Type of soil in 2020	0.896694	
Rabi Seasons	Type of soil in 2020	1.000000	

	Kharif Seasons	Type of soil in 2021	\
Kharif Seasons	Type of soil in 2022	1.000000	
Rabi Seasons	Type of soil in 2022	1.000000	
Rabi Seasons	Type of soil in 2021	1.000000	
Kharif Seasons	Type of soil in 2021	1.000000	
Kharif Seasons	Type of soil in 2020	0.896694	
Rabi Seasons	Type of soil in 2020	1.000000	

	Kharif Seasons	Type of soil in 2020	\
Kharif Seasons	Type of soil in 2022	0.896694	
Rabi Seasons	Type of soil in 2022	0.896694	
Rabi Seasons	Type of soil in 2021	0.896694	
Kharif Seasons	Type of soil in 2021	0.896694	
Kharif Seasons	Type of soil in 2020	1.000000	
Rabi Seasons	Type of soil in 2020	0.896694	

	Rabi Seasons	Type of soil in 2020
Kharif Seasons	Type of soil in 2022	1.000000
Rabi Seasons	Type of soil in 2022	1.000000
Rabi Seasons	Type of soil in 2021	1.000000
Kharif Seasons	Type of soil in 2021	1.000000
Kharif Seasons	Type of soil in 2020	0.896694
Rabi Seasons	Type of soil in 2020	1.000000

```
suspect_cols = [
    'Kharif Seasons Type of water bodies in hectares 2022',
    'Rabi Seasons Type of water bodies in hectares 2022',
    'Rabi Seasons Type of water bodies in hectares 2021',
    'Kharif Seasons Type of water bodies in hectares 2021',
    'Kharif Seasons Type of water bodies in hectares 2020',
    'Rabi Seasons Type of water bodies in hectares 2020'
]
```

```
string_correlation_matrix = df[suspect_cols].apply(lambda x: x.astype('category').cat.codes).c  
print(string_correlation_matrix)
```

Kharif Seasons Type of water bodies in h

ectares 2022 \

Kharif Seasons	Type of water bodies in hectare...
1.000000	
Rabi Seasons	Type of water bodies in hectares 2022
1.000000	
Rabi Seasons	Type of water bodies in hectares 2021
1.000000	
Kharif Seasons	Type of water bodies in hectares...
1.000000	
Kharif Seasons	Type of water bodies in hectares...
0.768974	
Rabi Seasons	Type of water bodies in hectares 2020
1.000000	

Rabi Seasons Type of water bodies in hect

ares 2022 \

Kharif Seasons	Type of water bodies in hectare...
1.000000	
Rabi Seasons	Type of water bodies in hectares 2022
1.000000	
Rabi Seasons	Type of water bodies in hectares 2021
1.000000	
Kharif Seasons	Type of water bodies in hectares...
1.000000	
Kharif Seasons	Type of water bodies in hectares...
0.768974	
Rabi Seasons	Type of water bodies in hectares 2020
1.000000	

Rabi Seasons Type of water bodies in hect

ares 2021 \

Kharif Seasons	Type of water bodies in hectare...
1.000000	
Rabi Seasons	Type of water bodies in hectares 2022
1.000000	
Rabi Seasons	Type of water bodies in hectares 2021
1.000000	
Kharif Seasons	Type of water bodies in hectares...
1.000000	
Kharif Seasons	Type of water bodies in hectares...
0.768974	
Rabi Seasons	Type of water bodies in hectares 2020
1.000000	

Kharif Seasons Type of water bodies in he

ctares 2021 \

Kharif Seasons	Type of water bodies in hectare...
1.000000	
Rabi Seasons	Type of water bodies in hectares 2022
1.000000	
Rabi Seasons	Type of water bodies in hectares 2021
1.000000	
Kharif Seasons	Type of water bodies in hectares...
1.000000	
Kharif Seasons	Type of water bodies in hectares...

0.768974
Rabi Seasons Type of water bodies in hectares 2020
1.000000

Kharif Seasons Type of water bodies in he

ctares 2020 \

Kharif Seasons	Type of water bodies in hectare...
0.768974	
Rabi Seasons	Type of water bodies in hectares 2022
0.768974	
Rabi Seasons	Type of water bodies in hectares 2021
0.768974	
Kharif Seasons	Type of water bodies in hectares...
0.768974	
Kharif Seasons	Type of water bodies in hectares...
1.000000	
Rabi Seasons	Type of water bodies in hectares 2020
0.768974	

Rabi Seasons Type of water bodies in hect

ares 2020

Kharif Seasons	Type of water bodies in hectare...
1.000000	
Rabi Seasons	Type of water bodies in hectares 2022
1.000000	
Rabi Seasons	Type of water bodies in hectares 2021
1.000000	
Kharif Seasons	Type of water bodies in hectares...
1.000000	
Kharif Seasons	Type of water bodies in hectares...
0.768974	
Rabi Seasons	Type of water bodies in hectares 2020
1.000000	